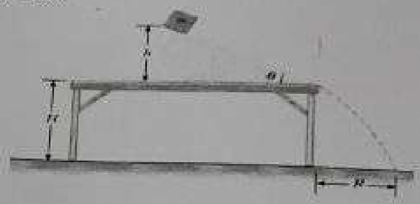
2. A block of mass $m = 2$ kg is released from rest at $h = 0.5$ m above the surface of a table, at the top of $\theta = 30^0$ incline as shown in Fig below. The frictionless incline is fixed on a table of height $H = 2$ m.

a) Determine the acceleration of the block as it slides down the incline.
b) What is the velocity of the block as it leaves the incline?
c) How far from the table will the block hit the floor?

(Use: $g = 10\ m/s^2$, $\sin 30^0 = 0.5$ & $\cos 30^0 = 0.87$)



Given

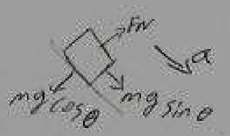
$m = 2kg$

$h = 0.5m$

$\theta = 30^0$

$H = 2m$

4



a)

$\Sigma F = 0$

$$\frac{mg\sin\theta}{m} = \frac{ma}{m}$$

$$\frac{\cancel{m}g\sin\theta}{\cancel{m}} = a$$

$10 \times 0.5 = a$ $\quad a = 5\ m/s^2$

$\Delta me = 0$

b) $\cancel{m}gh = \frac{1}{2}\cancel{m}v^2$

$v = \sqrt{2gh}$

$= \sqrt{2 \times 10 \times 0.5}$

$= \sqrt{10} = 3.16\ m/s$

~~$V = U + at$ $\quad U = 0$~~

$V = U + at$ $\quad U = 0$

$V = at$

$t = V/a = \frac{3.16}{5} = 0.632$

c) $R = Vt$

$= 3.16 \times 0.632$

$= 1.99 \approx 2m$

8

PART I: FILL IN THE BLANK SPACE ITEMS

Direction: **Fill in the blank space with the most simplified answers [8%, 2pts each]**

1. Let $\vec{A} = 2\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{B} = -2\hat{i} + \hat{j} - 2\hat{k}$. Find

   a) $\vec{A} \cdot \vec{B} =$ [redacted] -7

   b) $\vec{A} \times \vec{B} =$ [redacted] $-4\hat{i} + 4\hat{k}$

   c) A third vector of magnitude 10 units which is perpendicular to both $\vec{A}$ and $\vec{B}$ [redacted] $-5\sqrt{2}\hat{i} + 5\sqrt{2}\hat{k}$

   d) The area of a parallelogram with adjacent sides of $\vec{A}$ and $\vec{B}$ [redacted] $= \sqrt{32} = 4\sqrt{2}$

2. A 20 g bullet moving horizontally at 50 m/s strikes a 7 kg block resting on a table. The bullet embeds in the block after collision. The speed of the block after collision is [redacted] $V = 0.14\,\frac{m}{s}$

3. An object moves along a straight line. At a time t the distance x of an object from the origin is given by $x(t) = (40 + 12t - t^2)\,m$.

   a) What is the acceleration at $t = 2$sec [redacted] $a = -2\,\frac{m}{s^2}$

   b) How far would the object travel before coming rest? [redacted] $76\,m$

4. A mass of 0.5kg moving with speed of 1.5 m/s on a horizontal smooth surface collides with a massless spring of force constant 50N/m. The maximum compression of the spring would be [redacted] $x = 0.15\,m = 15\,cm$

PART II: WORKOUT ITEMS

Direction: Solve the following problems by showing all the necessary steps briefly only in the space provided [12%]

1. A ball is tossed from an upper-story window of a building. The ball is given an initial velocity of 25 m/s at an angle of 53^0 above the horizontal. It strikes the ground 3 seconds later.
   a) How far horizontally from the base of the building does the ball strike the ground?
   b) Find the height from which the ball was thrown.
   c) With what velocity it strikes the ground?
   (Use: $\sin 53^0 = 0.8$, $\cos 53^0 = 0.6$ & $g = 10\ m/s^2$)

2.5

Given
$U = 25\ m/s$
$\theta = 53^0$
$t = 3\ sec$

$h = y$

$U_x = U\cos\theta = 25 \times 0.6 = 15\ m/s$
$U_y = U\sin\theta = 25 \times 0.8 = 20\ m/s$

$V_x = V\cos\theta = 15\ m/s$
$V_y = U_y - gt$
$= 20 - (10 \times 3)$
$= -10\ m/s$

1.5

a, $R = V_x t$
$\cancel{= 15 \times 3 = 45}$
$= 15\ m/s \times 3s = 45m$

1

b, $Y = U_y t \pm \frac{1}{2} a t^2$ $\quad a = -g$
$= U_y t - \frac{1}{2} g t^2$
$= 20(3) - \frac{1}{2}(10 \times 3^2)$
$= 60 - 45 = 15m$

c $V = \sqrt{V_x^2 + V_y^2}$
$= \sqrt{(15^2) + (-10^2)}$
$= \sqrt{325} = 18.03\ m/s \approx 18\ m/s$

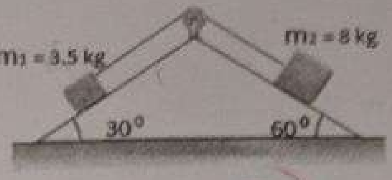
2. Two blocks of Masses $m_1 = 3.5$ kg and $m_2 = 8$ kg are connected by a massless string that passes over a frictionless pulley as shown in the figure below. The coefficient of friction between the incline and the block is 0.25.

a) Draw free body diagram.

b) The acceleration of the masses

c) The tension in the string.

(Use: $g = 10\ m/s^2$, $\sin 30^0 = 0.5$, $\cos 30^0 = 0.86$, $\sin 60^o = 0.86$, $\cos 60^o = 0.5$)



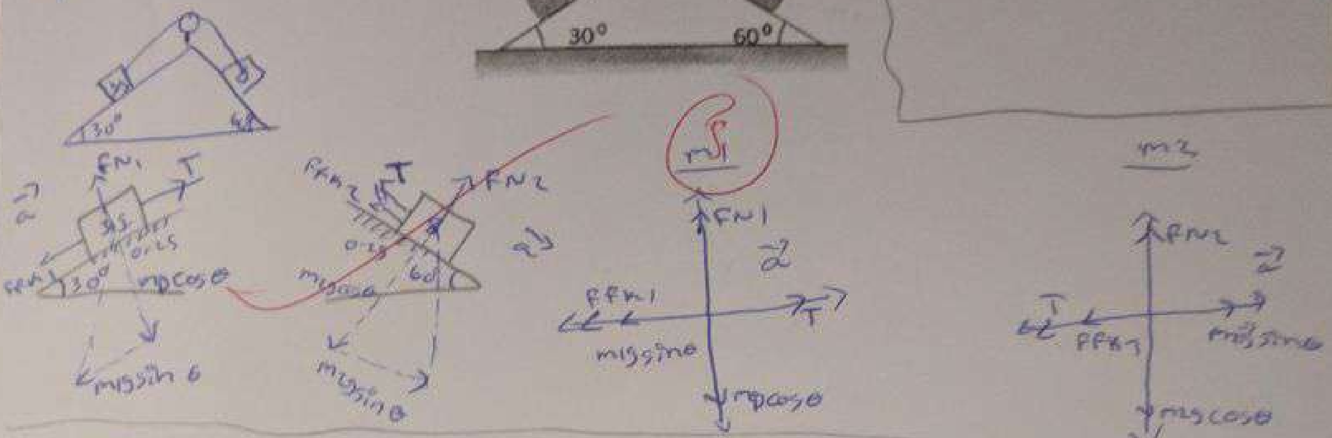
c) $T - m_1 g \cos\theta\, \mu_k - m_1 g \sin\theta = m_1 \vec{a}$

$T - 7.6 - 17.5 = 3.5 \times 2.97$

$T = 10.395 + 25.1$

$= 35.495\ N$ ##

a)



b) m_1

$\sum F_y = 0$

$F_{N1} - m_1 g \cos\theta = 0$

$F_{N1} = m_1 g \cos\theta$

$\sum F_x = m\vec{a}$

$T - F_{fk1} - m_1 g \sin\theta = m_1 \vec{a}$

$T - F_N \mu_k - m_1 g \sin\theta = m_1 \vec{a}$

$T - m_1 g \cos\theta\, \mu_k - m_1 g \sin\theta = m_1 \vec{a}$ --- ①

$\sum F_y = 0$

$F_{N2} - m_2 g \cos\theta = 0$

$F_{N2} = m_2 g \cos\theta$

m_2

$\sum F_x = m_2 \vec{a}$

$m_2 \vec{g} \sin\theta - T - F_{fk2} = m_2 \vec{a}$

$m_2 \vec{g} \sin\theta - T - F_{N2} \mu_k = m_2 \vec{a}$

$m_2 \vec{g} \sin\theta - T - m_2 g \cos\theta\, \mu_k = m_2 \vec{a}$ --- ①

T

$\begin{cases} \cancel{T} - m_1 g \cos\theta\, \mu_k - m_1 g \sin\theta = m_1 \vec{a} \\ m_2 \vec{g} \sin\theta - \cancel{T} - m_2 \vec{g} \cos\theta\, \mu_k = m_2 \vec{a} \end{cases}$

$m_2 \vec{g} \sin\theta - m_1 g \sin\theta - m_1 \vec{g} \cos\theta\, \mu_k - m_2 \vec{g} \cos\theta\, \mu_k = \vec{a}(m_1 + m_2)$

$8 \times 10 \times \sin 60^0 - 3.5 \times 10 \times \sin 30^0 - 3.5 \times 10 \times \cos 30^0 \times 0.25 - 8 \times 10 \times \cos 60 \times 0.25 = \vec{a}(11.5)$

$69.3 - 17.5 - 7.6 - 10 = \vec{a}(11.5)$

$\frac{34.2}{11.5} = \vec{a}$

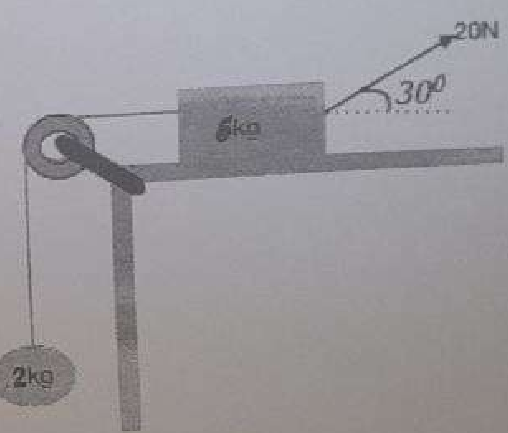
$\vec{a} = 2.97\ m/s^2$ ##

4

3. A block of mass 6kg on a rough, horizontal surface is connected to a ball of mass 2kg by a light-weight cord over a lightweight, frictionless pulley as shown in the figure below. A force of magnitude 20N at an angle 30^0 with the horizontal is applied to the block and the block slides to the right. The coefficient of kinetic friction between the block and the surface is 0.25.

a) Draw a free body diagram.
b) Find the acceleration of the objects.
c) Find the tension of a cord.

(Use: $g = 10\ m/s^2$, $\sin 30^0 = 0.5$ & $\cos 30^0 = 0.87$)



~~let~~ let $m_1 = 6kg$
$m_2 = 2kg$

Given
$m_1 = 6kg$
$m_2 = 2kg$
$\theta = 30^0$
$F = 20N$
$\mu_k = 0.25$

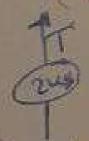
~~a) for~~ Soln

A. for mass m_1 or block of a body

where
f_f = friction force
T = Tension
F_N = normal force.
$F_x = F\cos\theta$ or horizontal component of F.
$F_y = F\sin\theta$ or vertical component of F.

0.5

for m_2 or ball.



0.5

b. for m_2 = $T - m_2g = m_2a$ 0.5
for m_1 = $F\cos\theta - T - f_f = m_1a$ 0.5
$f_f = \mu_k(m_1g - F\sin\theta)$

$= F\cos\theta - T - \mu_k(m_1g - F\sin\theta) = m_1a$
$T - m_2g = m_2a$ we add the two equation

$= F\cos\theta - m_2g - \mu_k(m_1g - F\sin\theta) = (m_1+m_2)a$

$= F\cos\theta - m_2g - \mu_k(m_1g - F\sin\theta)$

$a = \dfrac{F\cos\theta - m_2g - \mu_k(m_1g - F\sin\theta)}{m_1+m_2}$ 0.5

$= \dfrac{(20N)(0.87) - (2kg)(10m/s^2) - (0.25)\left((6kg)(10m/s^2) - (20N)(0.5)\right)}{6kg + 2kg}$

$\dfrac{-15.1}{8kg} = -1.88$
$\boxed{= -1.8875 m/s^2}$

c. $T - m_2g = m_2a$
$T = m_2a + m_2g$
~~$= (2kg)(-6.575) + (2kg)(10m/s^2)$~~
~~$= -13.15N + 20N$~~
~~$= 6.85N$~~

$T = m_2a + m_2g$
$= (2kg)(-1.8875) + (20N)$
$= -3.775 + 20N$
$\boxed{T = 16.225N}$

1

PART I: FILL IN THE BLANK SPACE ITEMS

Direction: Fill in the blank space with the most simplified answers [6%, 1pt for each blank space]

1. For the three vectors, $\boldsymbol{A} = 3\hat{\imath} + 4\hat{\jmath} + 5\hat{k}$ and $\boldsymbol{B} = 6\hat{\imath} + 8\hat{\jmath} + 10\hat{k}$ find
   a) $\boldsymbol{A} \cdot \boldsymbol{B}$ = 100 units
   b) A third vector having magnitude 4 times $\boldsymbol{B}$ and a direction perpendicular to both vectors $\boldsymbol{A}$ & $\boldsymbol{B}$ = $0\hat{\imath} + 0\hat{\jmath} + 0k$ = 0 vector
2. An airplane moving horizontally at a height of 500m releases a heavy bomb to hit a target at a horizontal distance of 1000m. The initial velocity of the plane to hit the target is 100 m/s.
3. The coordinates of a bird flying in the xy-plane are given by $x(t) = at + b$ and $y(t) = ct^2 + d$, where $a = 1.00\ m/sec$, $b = 1.00\ m$, $c = 0.125 \frac{m}{sec^2}$, and $d = 1.00\ m$
   a) The speed of the bird at t = 5.0s is 1.4 m/s.
   b) The average velocity between t = 1 sec and t = 3sec = ~~[illegible]~~ 6m/s $\hat{\imath}$ + 4.125 m/s $\hat{\jmath}$
   c) The acceleration of the bird for the time interval t = 0 to t = 10s is $\boxed{0\hat{\imath} + 0.25\hat{\jmath}}$ m/s²

PART II: SHORT ANSWER ITEMS

Direction: Give short answers for the following questions only in the space provided [4%, 1.5pt each]

1. Why are action and reaction forces not added to give a zero resultant force equal in magnitude and opposite direction?
   - b/c as the law said for every action their is always equal but opposite force of reaction. the person is ~~not~~ only exerting force on one point

2. An object moves in a circular path with constant speed v.
   a) Is the velocity of the object constant? Explain
   - The magnitude of the velocity is constant $\boxed{\text{but}}$ since velocity has direction when it's in circular path it's not constant because there is change in direction
   b) Is its acceleration constant? Explain
   - The magnitude of $\vec{a}$ is constant but this one also has direction ~~[illegible]~~ hence $\vec{a}$ is always nonconstant in a circular path

3. State the following Laws.
   a) Newton's 1st law of motion states that "on the absence of unbalanced force a body at rest will remain at rest a body in motion will remain in motion with constant velocity $\vec{v}$"
   b) Newton's 2nd law of motion

States that "The sum of all ~~net~~ the forces ($\vec{F}_{net}$) is directly proportional to the acceleration"

$\vec{F}_{net} \propto \vec{a}$

$\vec{F}_{net} = (\text{constant})\,\vec{a}$

$\boxed{\vec{F}_{net} = m\vec{a}}$

(t, 0.25t)

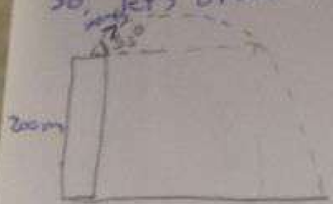
$\frac{(10, 2.5) - (0)}{10 - 0}$

PART III: WORKOUT ITEMS

Direction: Solve the following problems by showing all the necessary steps briefly only in the space provided [10%, 5points for each question]

1. A man throws a ball upward from the top of a building 200 m above the ground level with an initial speed of 100 m/s at an angle of 53° above the horizontal. Calculate
   a) How long does it take the ball to reach the ground?
   b) What is the speed of the ball just before it strikes the ground?
   c) The horizontal distance traveled from the base of the building
   (Use: $\sin 53° = 0.8$, $\cos 53° = 0.6$, $g = 10\ m/s^2$)

So, let's draw the diagram



$v_i = 100 m/s$ $\quad$ a, t=?

$\Delta y = 200 m$ $\quad$ b, v=?

$\theta = 53°$ $\quad$ c, Δx=?

$a_x = 0$ $\quad$ $a_y = -g$

5

(a) $\Delta y = v_i t + \frac{1}{2} a t^2$

$\Delta y = v_{iy} t + \frac{1}{2} a_y t^2$

$\Delta y = v_i \sin\theta t - \frac{1}{2} g t^2$

$y_f - y_i = v_i \sin\theta t - \frac{1}{2} g t^2$

$y_f = y_i + v_i \sin\theta t - \frac{1}{2} g t^2$, $y_f = 0$

$0 = 200m + 100 \times \sin 53° \times t - \frac{1}{2} \times 10 \times t^2$

$0 = 200m + 100 \times 0.8 \times t - 5t^2$

$0 = 200 + 80t - 5t^2 \Rightarrow 5t^2 - 80t - 200 = 0$

$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{80 \pm \sqrt{6400 + 4000}}{10}$

$t = 18.19 s$ ✓, $t = -2.19 s$ ✗

$\boxed{t = 18.19 s}$ #

(c) $\Delta x = v_i t + \frac{1}{2} a t^2$, $a_x = 0$

$\Delta x = v_{ix} t + \frac{1}{2} a_x t^2$

$\Delta x = v_i \cos\theta t$

$= 100 \times \cos 53° \times 18.19 s$

$= 100 \times 0.6 \times 18.19 s$

$= 60 \times 18.19$

$= 1091.4 m/s$ #

(b) $V = \sqrt{v_x^2 + v_y^2}$

$v_x = v_i \cos\theta = 100 \times \cos 53° = 100 \times 0.6 = 60 m/s$

$v_y = v_i \sin\theta - gt = 100 \times \sin 53° - 10 \times (18.19 s)$

$= 100 \times 0.8 - 181.9 s$

$= 80 - 181.9 s$

$= 101.9 m/s$

$V = \sqrt{(60 m/s)^2 + (101.9 m/s)^2}$

$= \sqrt{3600 + 10,383.61}$

$= \sqrt{13,983.61}$

$= 118.25 m/s$ #